

Alessandro Guenna

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I am a young mechanical and energy engineer with a strong passion for the development of innovative solutions in the context of the green transition. Confident in the use of Python and MATLAB, which I practice in personal projects showcased on my portfolio website.

Education

Mälardalen University (MDU), Västerås, Sweden - Master's thesis internship Aug 2026 – present

- Techno-economic study on waste heat recovery in the steel industry
- Developing a MATLAB model for different plant layouts based on Organic Rankine Cycle and District Heating, simulating design and off-design performance
- Assessing energy recovery, payback period, LCOE and CO₂ savings

University of Genoa, Genoa, Italy - Master's degree in Energy Engineering 2024 – present

University of Genoa, Genoa, Italy - Bachelor's degree in Mechanical Engineering 2020 – 2024

- Final grade 103/110
- Thesis: experimental evaluation of the performance of a magnetic refrigerator prototype, in collaboration with the company Bi-Freezer s.r.l.

University of Seville, Seville, Spain - Erasmus BIP "Energy communities for smart cities" February 2026

- Technical analysis for optimal sizing of PV plants based on load demand
- Business and regulatory assessment of REC projects

Projects

Winner of "Energy and Innovation – From Knowledge to Entrepreneurship", Wylab news
organized by Wylab and TICASS

- Competition aimed at developing managerial and entrepreneurial skills among UNIGE students, focused on innovative solutions for the energy sector
- Designed a business model for an industrial plant converting plastic waste into liquid fuel via catalytic pyrolysis (suitable for Otto cycle engines), promoting circular economy

Technical skills

Programming languages: Python, Excel VBA for automated worksheets

Software:

- MATLAB and Simulink
- Ansys Fluent and CFX
- Autodesk Inventor
- HOMER Pro
- Microsoft Office

Languages

Italian (native), English (advanced), German (basic)